

Proposed Customized Automated Revit Toolset

Enhancing Productivity in Autodesk Revit

Presented by: Biswambar Kshetri
Department: REC RDK Building Advisory and Consultancy
Date: 03 August 2026

What We'll Cover

01

Introduction & toolset Overview

Ribbon Detail

02

Script 1 — Audit Tools

Apply View Template ,View Template Check – Sheet

03

Script 2 — Automatic Dimension Tool

Auto dimension Wall, Floor, Foundation and Column

04

Script 3 — Automatic Export & Schedule Utilities

Export Multiple Schedule to Excel and CSV

05

Script 4 — Placement of Object By Co-ordinates

Geometry Placed By XYZ coordinates

06

Script 3 -Reassign Level

Reassign Level Without move element or change

07

Script 3 –Sheet Exporter Pro+

Export Sheet, PDF,DWG, DGN, IFC, Image with Custom File Name Same as RTV Exporter Tools

08

Script 3 –Worksets Manager

Create Worksets , Rename , Delete and Assign element into worksets

09

Script 9 — Auto Room Dimension

Dimension Internal room wall to wall Vertical & Horizontal

10

Script 10 — Selection Filter by Type

Filter Element by Type inside Revit Not Possible

11

Script 11 — Pile and Column Coordinate Writer

Write XYZ Coordinate in Parameter in Revit Pile and Column

12

Script 12— Covert Model in place to Generic

Convert Model in place to Generic Family and Place same location

13

Script 13— Join/Disjoin

Join and Disjoin Wall end and Structural Framing

14

Script 14 –Auto place door

Place Door by click two point -Useful Point Cloud Project

15

Script 15 –Auto place Windows

Place Windows by Two Point -Useful Point Cloud Project

16

Script 16 –Auto Create Wall Type

Create wall Type Click Two Point- change-Useful Point Cloud Project

17

Script 9 — QA Checklist Manual in Sheet

Place Manual Checklist in Sheet for Manual Check

18

Script 10 — Create Wall, Floor, Roof, Ceiling By Room

Model wall Floor Roof, Ceiling by Selecting Room-Useful Point Cloud Project

19

Script 11 — BIM Tracker Dashboard

This Tools Project Specific -St Pancras International Scan To BIM

20

Script 12— Clear BIM Status

This Tools Project Specific -St Pancras International Scan To BIM

21

Script 13— Combine Generic

Combine Generic Model

22

Script 14 –Tolerance Check 3D View Creator

This Tools Project Specific -St Pancras International Scan To BIM

23

Script 15 –Set RAM BIM Status

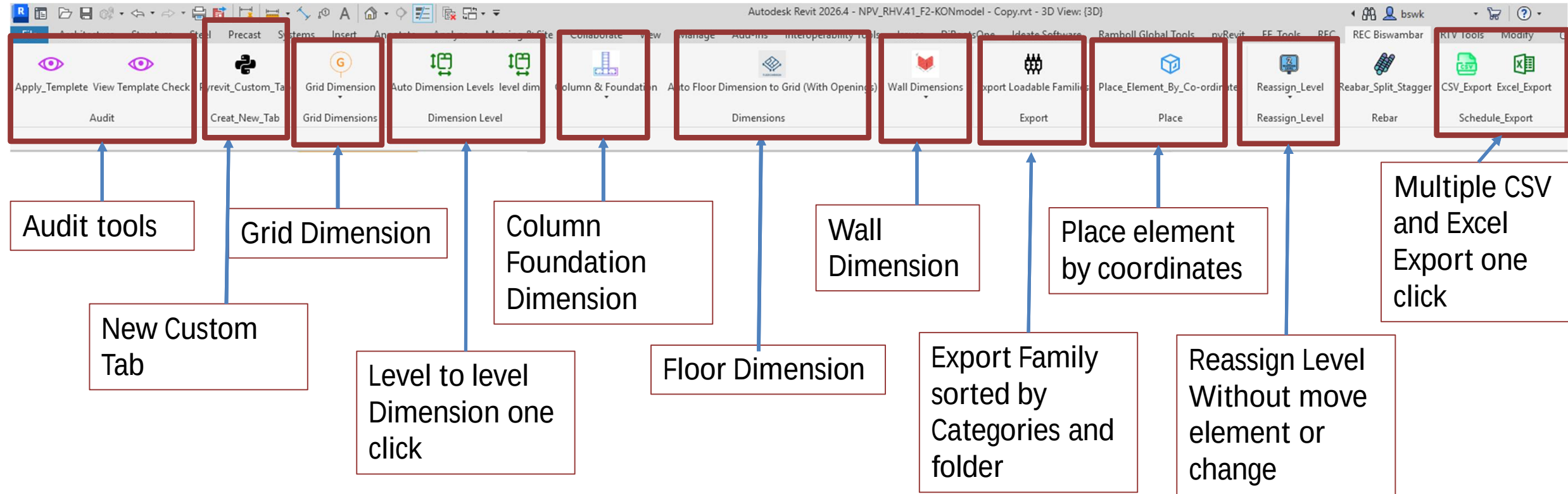
This Tools Project Specific -St Pancras International Scan To BIM

24

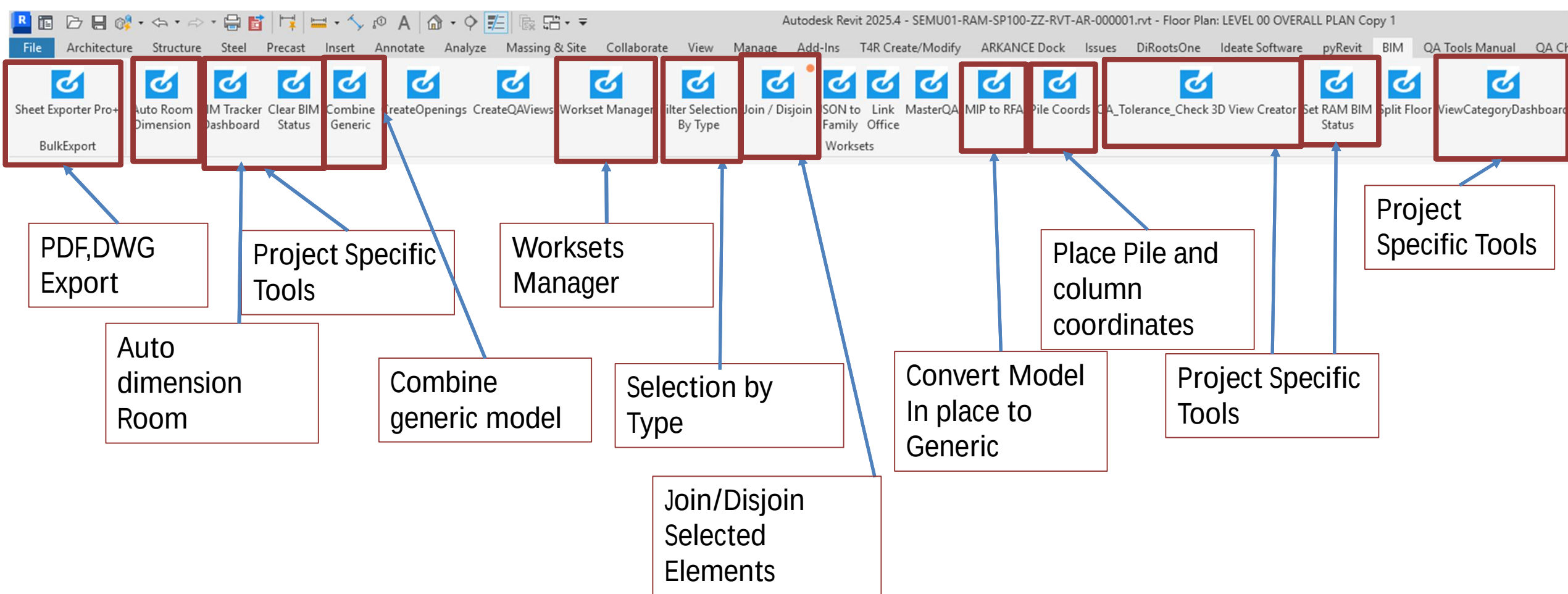
Script 16 –View Categories Dashboard

This Tools Project Specific -St Pancras International Scan To BIM

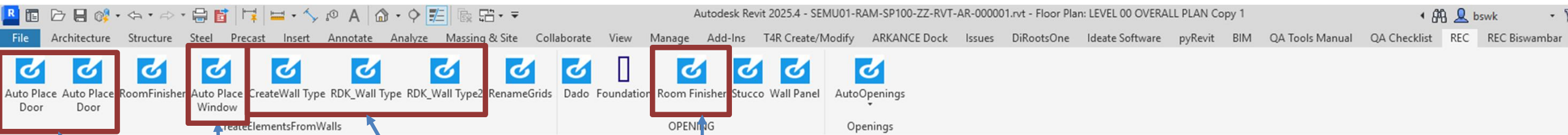
The REC Biswambar Ribbon



The BIM Ribbon



The REC Ribbon



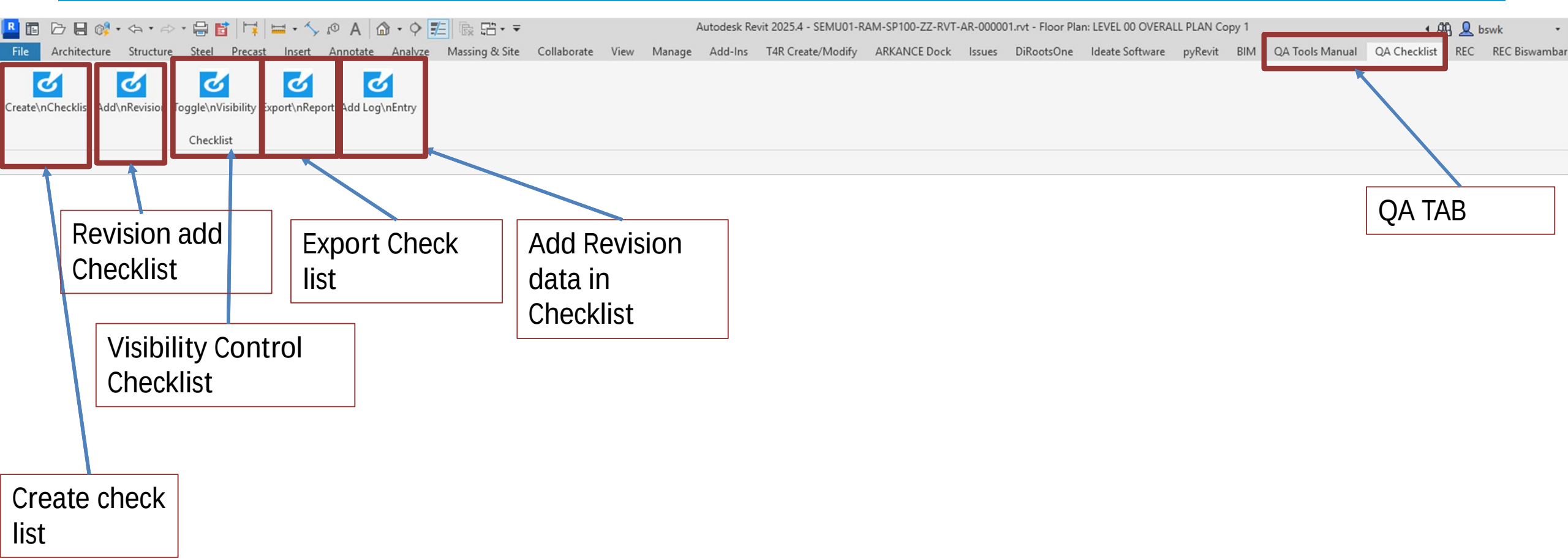
Door Place
Two Point

Window Place
Two Point

Wall Type 2
points

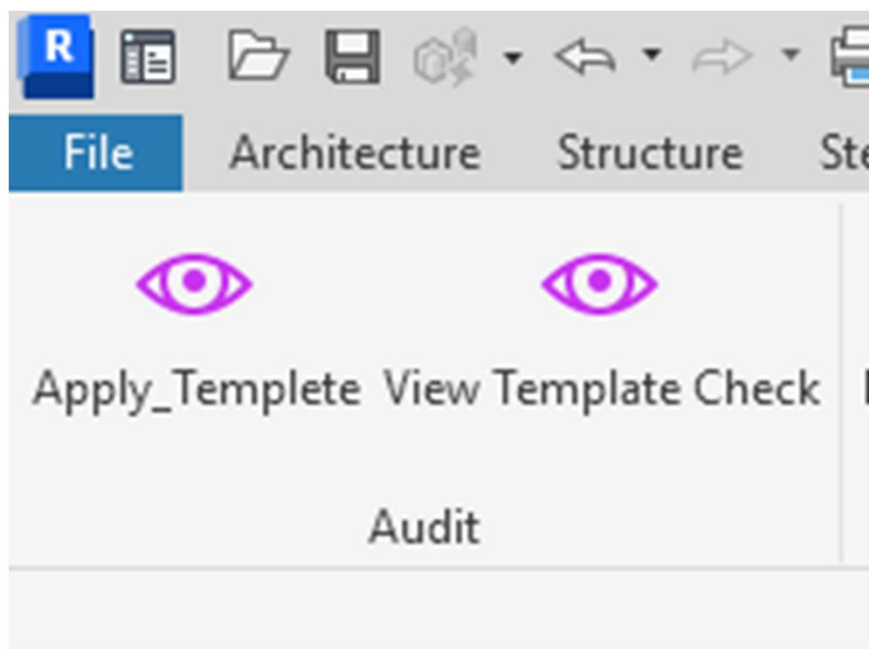
Model Wall , Floor,
Ceiling, Roof and Slope
Roof By Pick room

The QA Tools Manual Ribbon



Audit Tools

- Apply Template – Apply standard view templates in bulk
- View Template Check – Validate template usage across views
- Ensures consistent graphics and discipline standards



View_Template_Check					
NPV_RHV.41A8_M96_H1_EXX_N9001	KON-Disponering - Bygning A+B - Plan03-05	B1 - Plan 04 Copy 1	✓ Applied	MKYA_Displaner	
NPV_RHV.41C_M96_H1_E00_N9001	KON-Disponering - Bygning C - Plan00	Disp - Planer	✗ NO TEMPLATE		
NPV_RHV.41C_M96_H1_E00_N9001	KON-Disponering - Bygning C - Plan00	B2 - Plan 00 Copy 1	✓ Applied	MKYA_Displaner	
NPV_RHV.41C_M96_H1_E01_N9001	KON-Disponering - Bygning C - Plan01	Disp - Planer	✗ NO TEMPLATE		
NPV_RHV.41C_M96_H1_E01_N9001	KON-Disponering - Bygning C - Plan01	B2 - Plan 01 Copy 1	✓ Applied	MKYA_Displaner	
NPV_RHV.41C_M96_H1_E02_N9001	KON-Disponering - Bygning C - Plan02	Disp - Planer	✗ NO TEMPLATE		
NPV_RHV.41C_M96_H1_E02_N9001	KON-Disponering - Bygning C - Plan02	B2 - Plan 02 Copy 1	✓ Applied	MKYA_Displaner	
NPV_RHV.41D_M96_H1_E00_N9001	KON-Disponering - Bygning D - Plan00	Disp - Planer	✗ NO TEMPLATE		
NPV_RHV.41D_M96_H1_E00_N9001	KON-Disponering - Bygning D - Plan00	B3 - Plan 00 Copy 1	✓ Applied	MKYA_Displaner	
NPV_RHV.41E_M96_H1_E00_N9001	KON-Disponering - Bygning E - Plan00	Disp - Planer	✗ NO TEMPLATE		
NPV_RHV.41E_M96_H1_E00_N9001	KON-Disponering - Bygning E - Plan00	B4 - Plan 00 Copy 1	✓ Applied	MKYA_Displaner	
NPV_RHV.80x_M65_H5_EXX_N5001	KON Fundamentsdetaljer	SEDN_Fundamentsskitser	✓ Applied	03. Sheet Views - Plan-Floor	
NPV_RHV.80x_M65_H5_EXX_N5001	KON Fundamentsdetaljer	SEDN_Fundamentsskitser Side 2	✓ Applied	03. Sheet Views - Plan-Floor	
NPV_RHV.80x_M65_H5_EXX_N5002	KON Principsnit	SEDN_Fundamentsskitser Side 4	✓ Applied	03. Sheet Views - Plan-Floor	
00.00.01	Section 1	Section 1	✗ NO TEMPLATE		
00.00.02	Section 2	Section 2	✗ NO TEMPLATE		

Audit Summary

Total Sheets Scanned: 47
Total Views Checked: 49

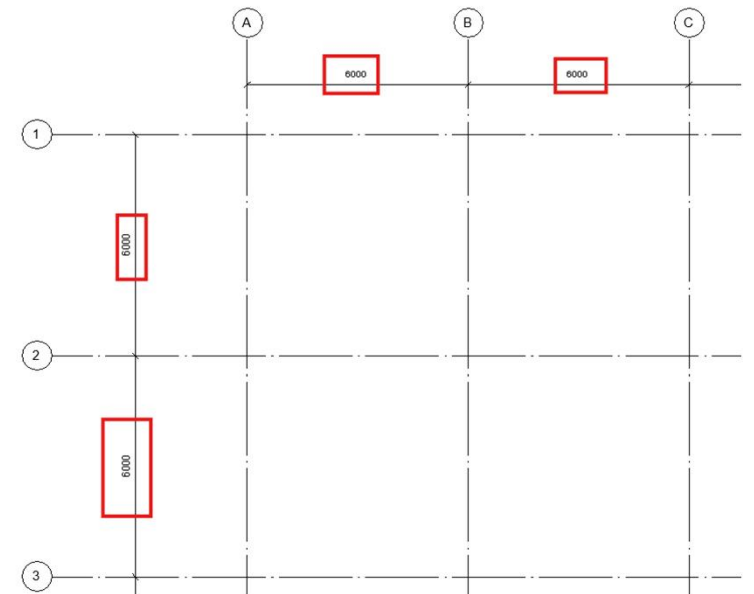
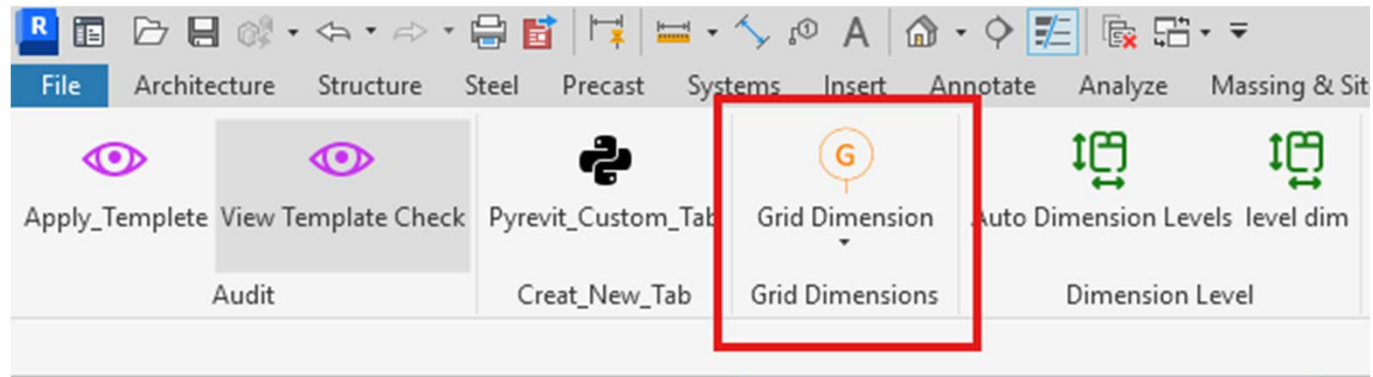
Template Usage Breakdown

Template: <None> | Count: 23
Template: MKYA_Floor Plan | Count: 9
Template: MKYA_Displaner | Count: 9
Template: MKYA_MV_Snit | Count: 5
Template: 03. Sheet Views - Plan-Floor | Count: 3

Overall Template Coverage: 53.1%

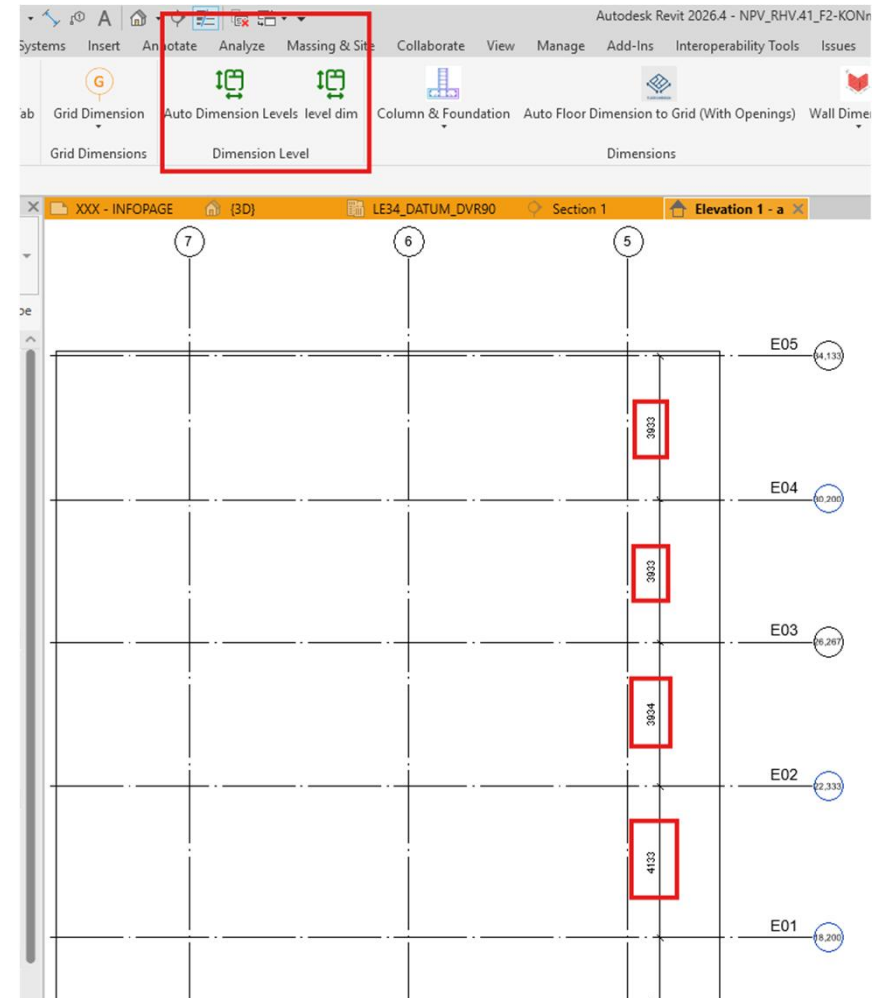
Grid Dimensioning

- Grid Dimension – Auto-create dimension strings along grids
- Eliminates manual click-by-click dimensioning
- Ensures consistent dimension placement



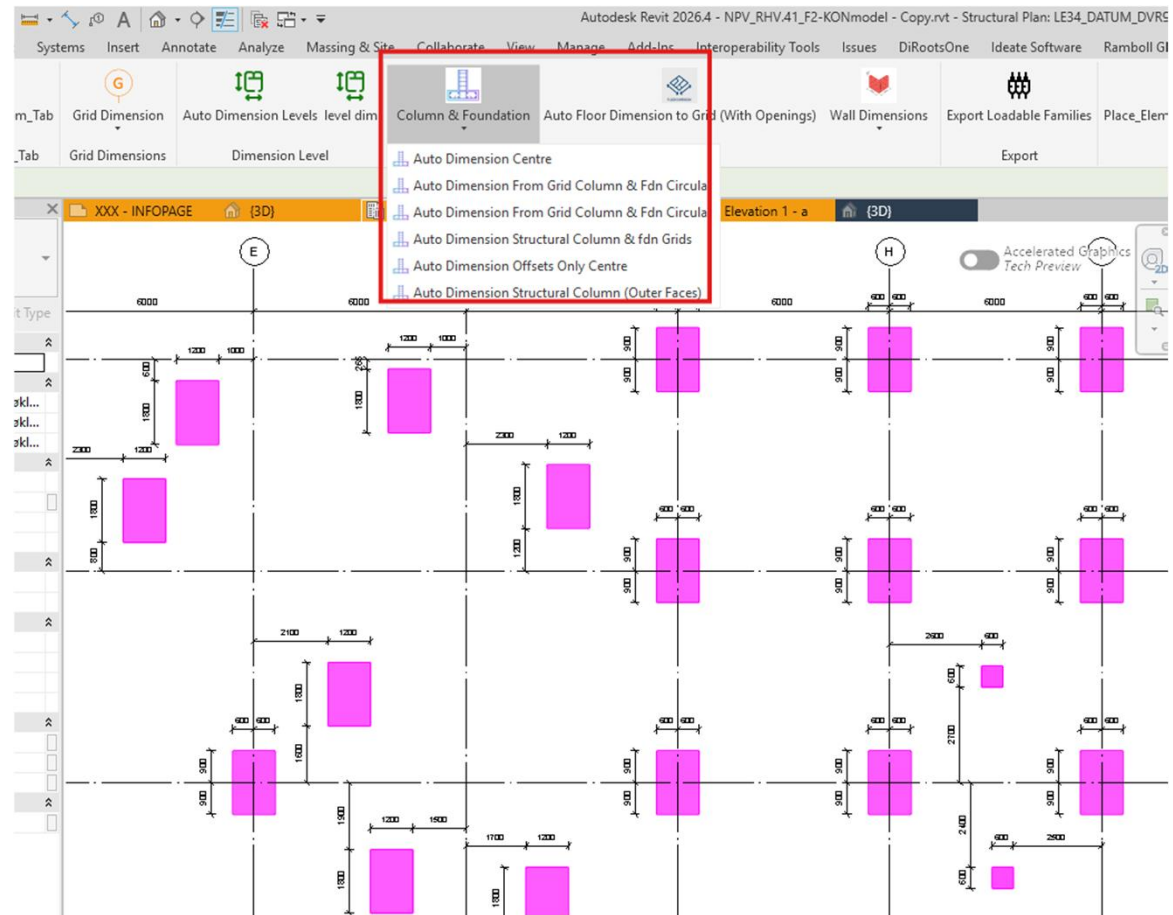
Level Dimensioning

- Auto Dimension Levels – Automatically dimension levels in sections/elevations
- Level Dim – Quick level-to-level dimension tool
- Saves hours in section drawing production



Smart Element Dimensioning

- Column & Foundation – Auto-dimension columns and footings to grids
- Auto Floor Dimension to Grid (with Openings) – Dimension slab edges and openings
- Wall Dimensions – Automated wall dimensioning



Smart Element Dimensioning- Workflow

Autodesk Revit 2026.4 - NPV_RHV.41_F2-KONmodel - Copy.rvt - Structural Plan: LE34_DATUM_DVR90

1 REC Biswambar

2 Wall Dimensions

3 Multiple

4 Select Plan Direction

5 Finish

6 View Range

7 Success

Plan view Wall Dimension generated perfectly with openings! Author-Biswambar Kshetri

pyRevit 6.4.0.26100+0515

Horizontal Chain (Left-to-Right across Vertical Grids)
Vertical Chain (Up-and-Down across Horizontal Grids)

Properties: Structural Plan: LE34_DATUM_DVR90

Graphics: View Scale: 1:100, Scale Value: 100, Display Model: Normal, Detail Level: Fine, Parts Visibility: Show Original, Visibility/Graphics Overrides: Edit..., Graphic Display Options: Edit..., Orientation: Project North, Wall Join Display: Clean all wall joins, Discipline: Coordination, Show Hidden Lines: By Discipline, Color Scheme Location: Background, Color Scheme: <none>, System Color Schemes: Edit..., Default Analysis Display: None, Sun Path: []

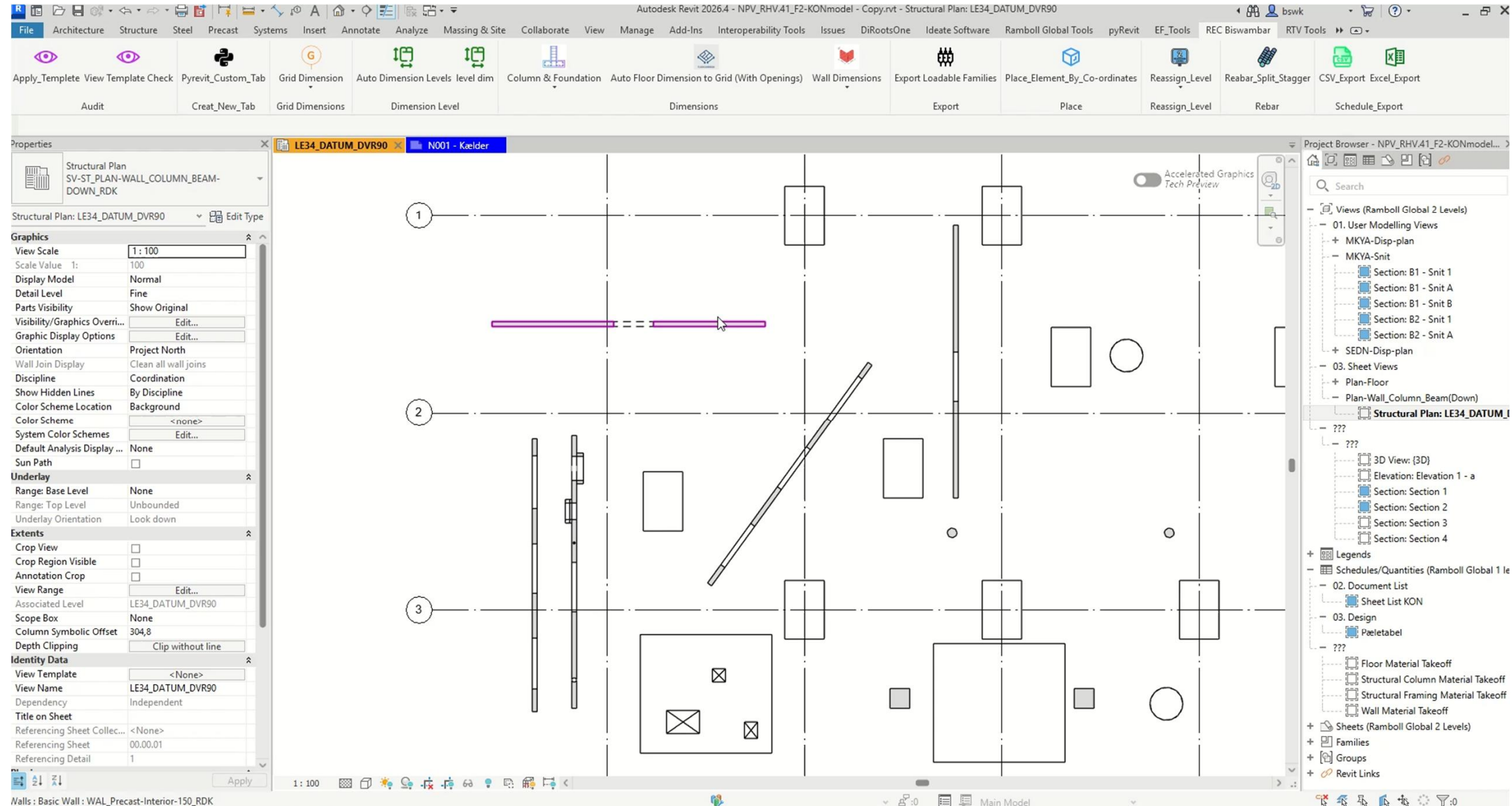
Underlay: Range: Base Level: None, Range: Top Level: Unbounded, Underlay Orientation: Look down

Extents: Crop View: [], Crop Region Visible: [], Annotation Crop: [], View Range: Edit..., Associated Level: LE34_DATUM_DVR90, Scope Box: [], Column Symbolic Offset: 304.8, Depth Clipping: Clip without line

Identity Data: View Template: <None>, View Name: LE34_DATUM_DVR90, Dependency: Independent, Title on Sheet: [], Referencing Sheet Collection: <None>, Referencing Sheet: 00.00.01, Referencing Detail: 1

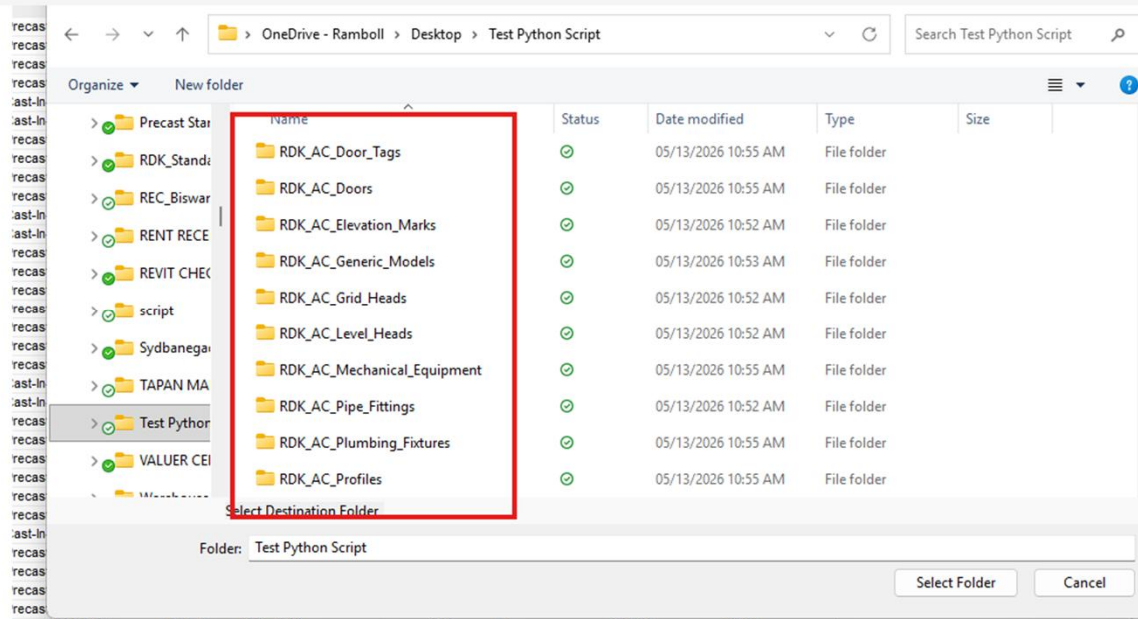
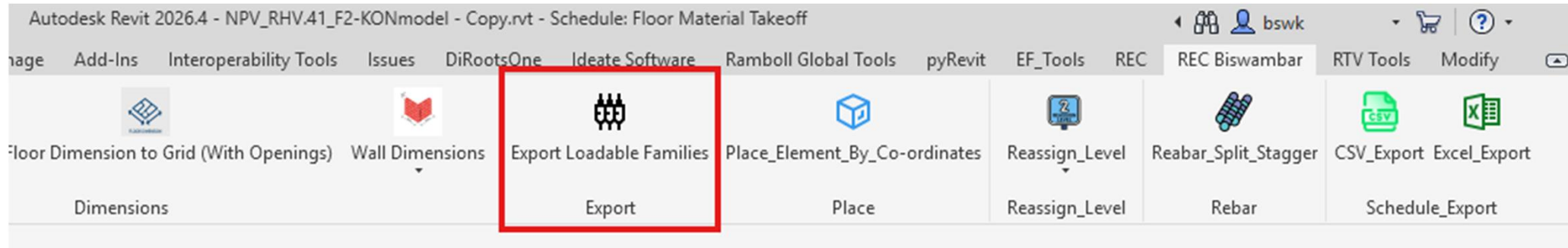
Project Browser: Views (Ramboll Global 2 Levels)
01. User Modelling Views
+ MKYA-Disp-plan
- MKYA-Snit
Section: B1 - Snit 1
Section: B1 - Snit A
Section: B1 - Snit B
Section: B2 - Snit 1
Section: B2 - Snit A
+ SEDN-Disp-plan
03. Sheet Views
+ Plan-Floor
Plan-Wall_Column_Beam(Down)
Structural Plan: LE34_DATUM_D

Smart Element Dimensioning- Video



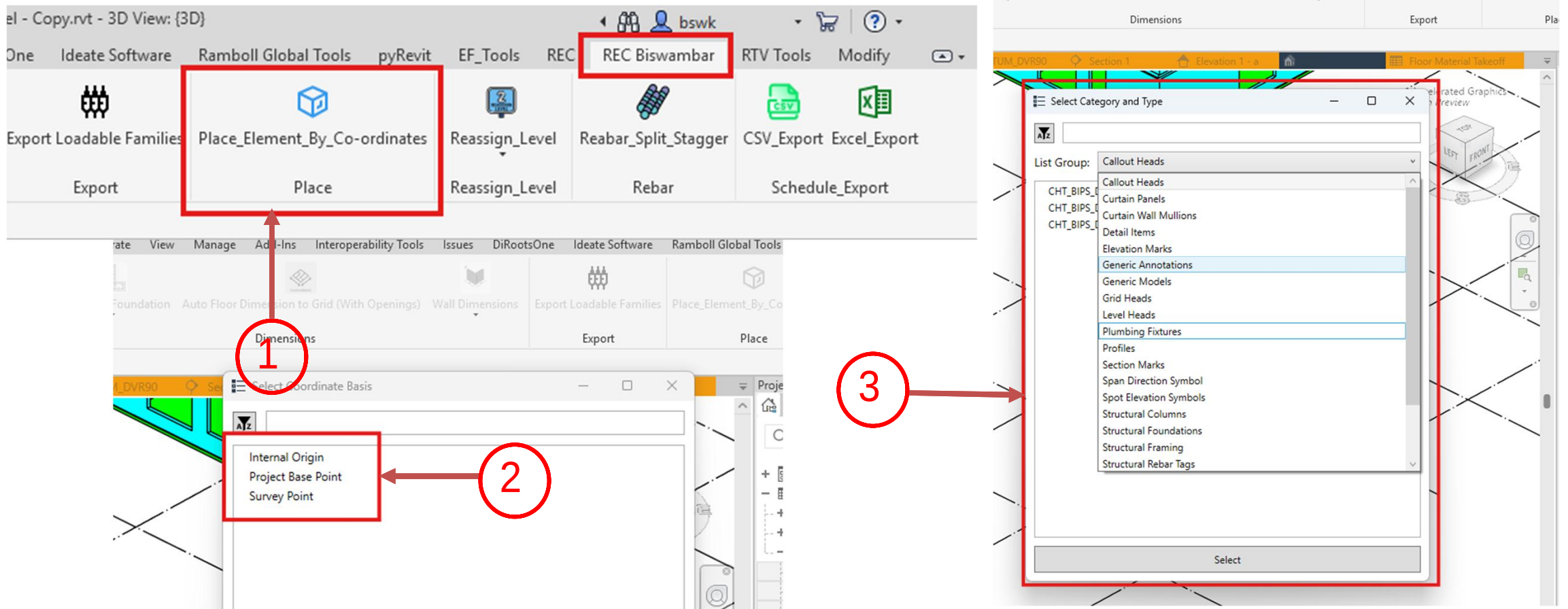
Export Panel

- Export Loadable Families – Batch-export all loadable families from the project
- Useful for family library management and project handover



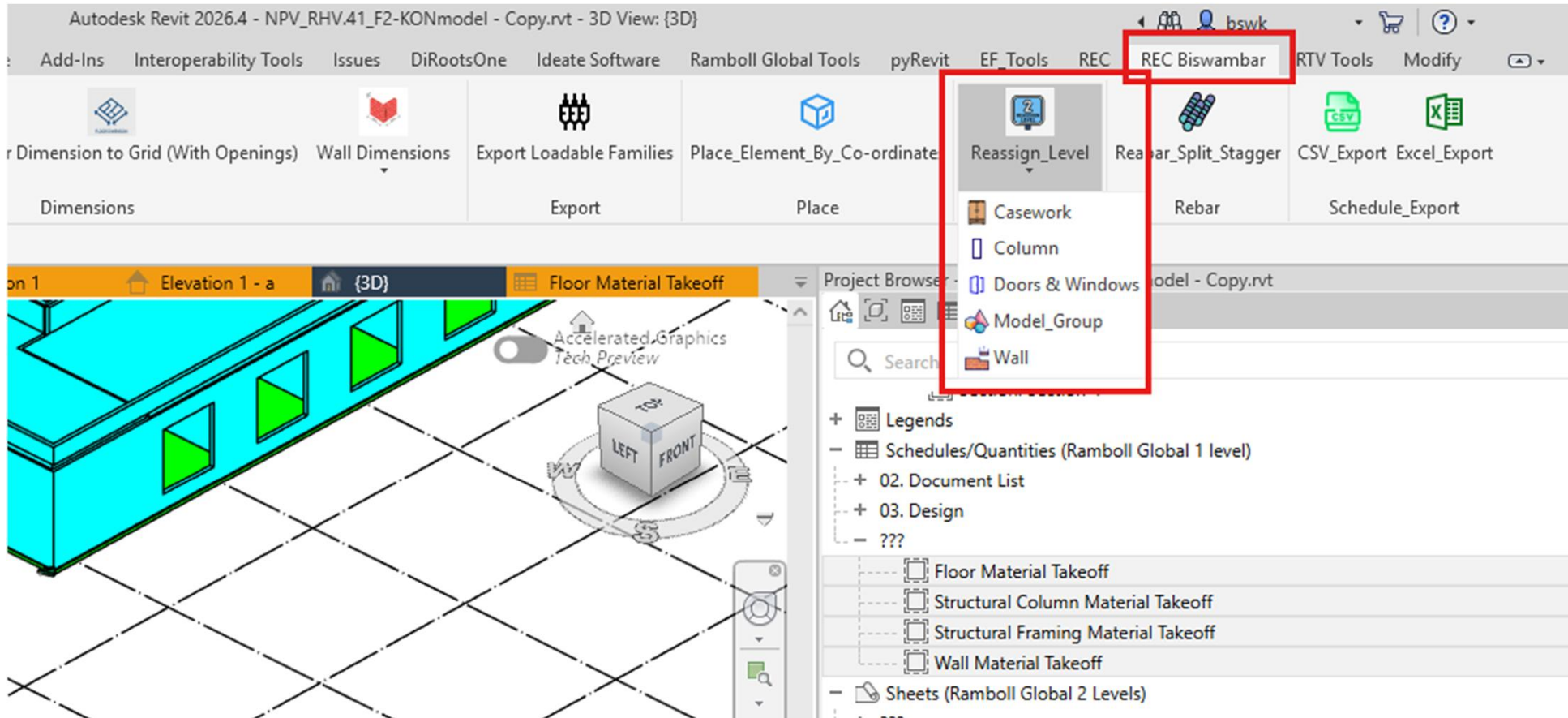
Coordinate-Based Placement

- Place Element by Co-ordinates – Place elements using X, Y, Z
- Useful for survey points, MEP fixtures, or pile layouts
- Reduces manual placement errors



Level Management

- Reassign Level – Move elements between levels while preserving their position
- Fixes common modeling errors quickly
- Prevents geometry shift during level changes



Data Export

- CSV Export – Export schedules to CSV format
- Excel Export – Direct export to Excel with formatting preserved
- Enables data analysis, BOQ preparation, and reporting

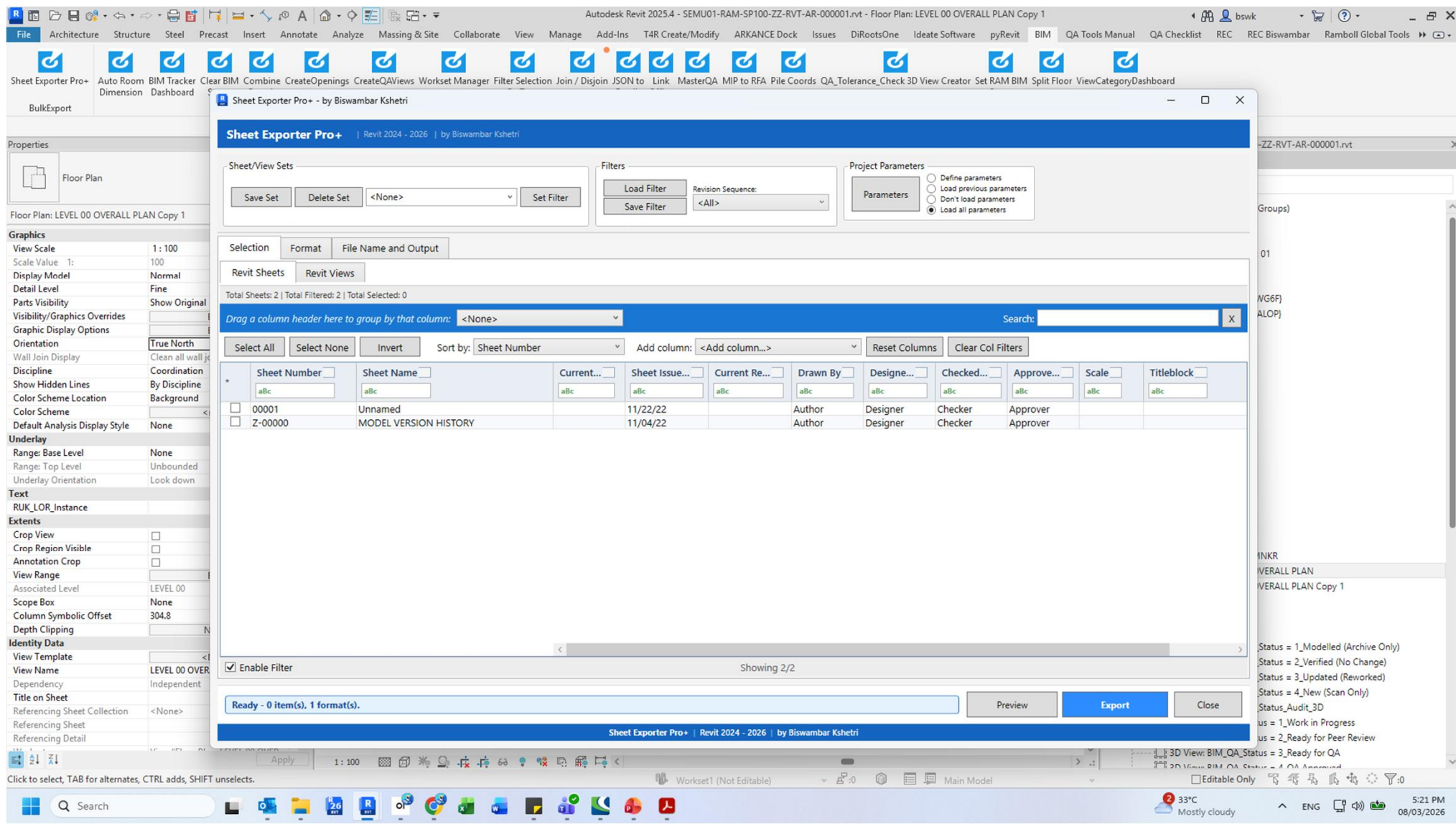
The screenshot displays the Autodesk Revit 2026.4 interface. The top ribbon shows the 'RTV Tools' tab, which includes 'CSV Export' and 'Excel Export' buttons, both highlighted with a red box. The main window shows a schedule titled '<Floor Material Takeoff>' with columns A through F. The data table is as follows:

A	B	C	D	E	F
Family and Type	Type	Default Thickness (	Count	Area (m²)	Material: Volume (m
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	46,14	12,46
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	36,10	11,55
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	629,24	201,36
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	253,30	68,39
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	313,49	37,62
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	209,66	25,16
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	81,40	26,05
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	573,78	183,61
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	81,40	26,05
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	573,78	183,61
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	126,99	15,24
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	97,56	11,71
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	81,40	26,05
Floor: FLO_Precast-320_RDK	FLO_Precast-320_RDK	320	1	573,78	183,61
Floor: FLO_Precast-180_RDK	FLO_Precast-180_RDK	180	1	93,23	16,78
Floor: FLO_Precast-180_RDK	FLO_Precast-180_RDK	180	1	608,04	109,45
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	143,81	38,83
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	107,44	29,01
Floor: FLO_Precast-270_RDK	FLO_Precast-270_RDK	270	1	320,79	86,61
Floor: FLO_Cast-In-Place-120_RDK	FLO_Cast-In-Place-120_RDK	120	1	545,89	65,51

The Project Browser on the right shows the 'Floor Material Takeoff' schedule selected under 'Schedules/Quantities (Ramboll Global 1 level)'.

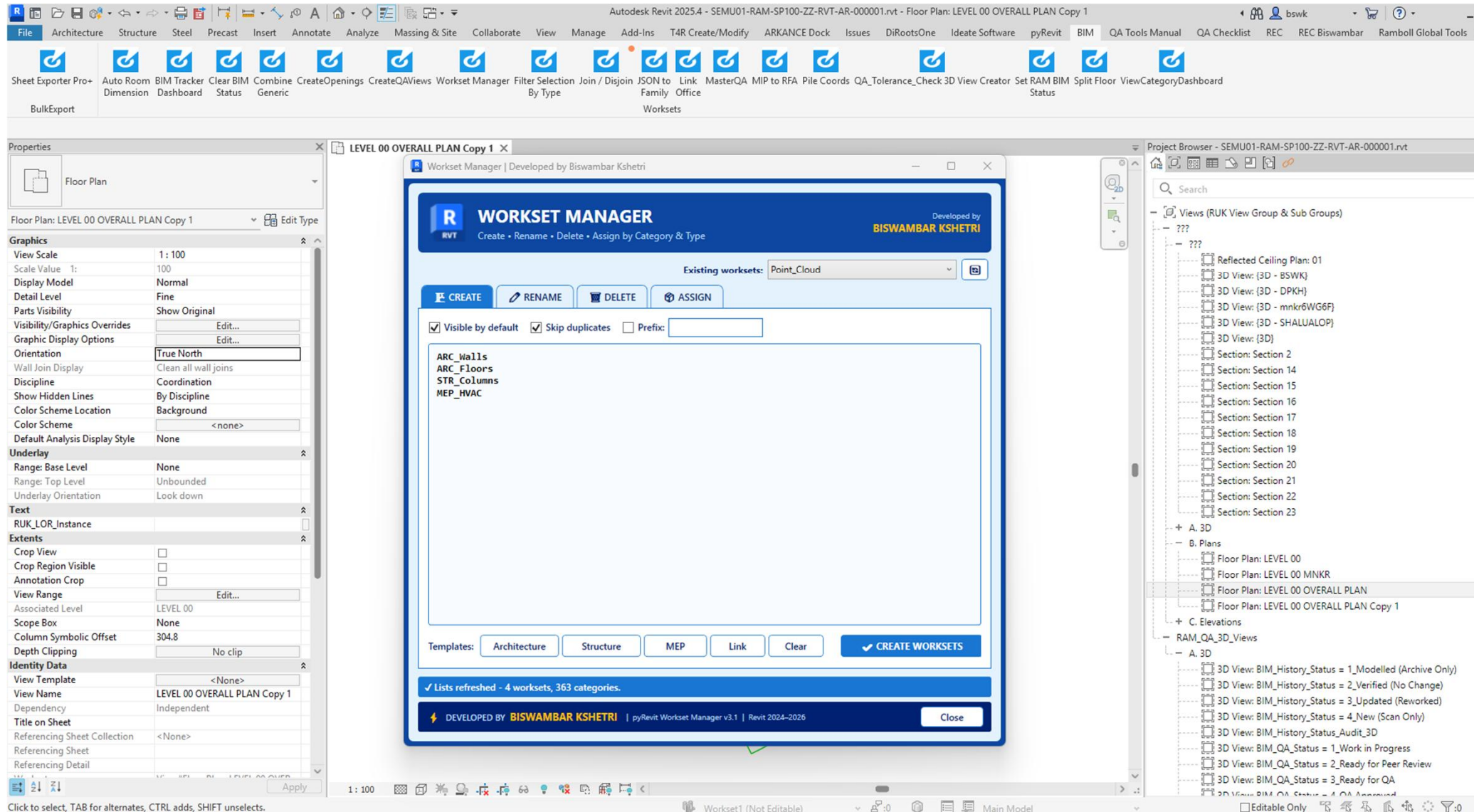
Sheet Exporter Pro+

Export Sheet, PDF,DWG, DGN, IFC, Image with Custom File Name Same as RTV Exporter Tools



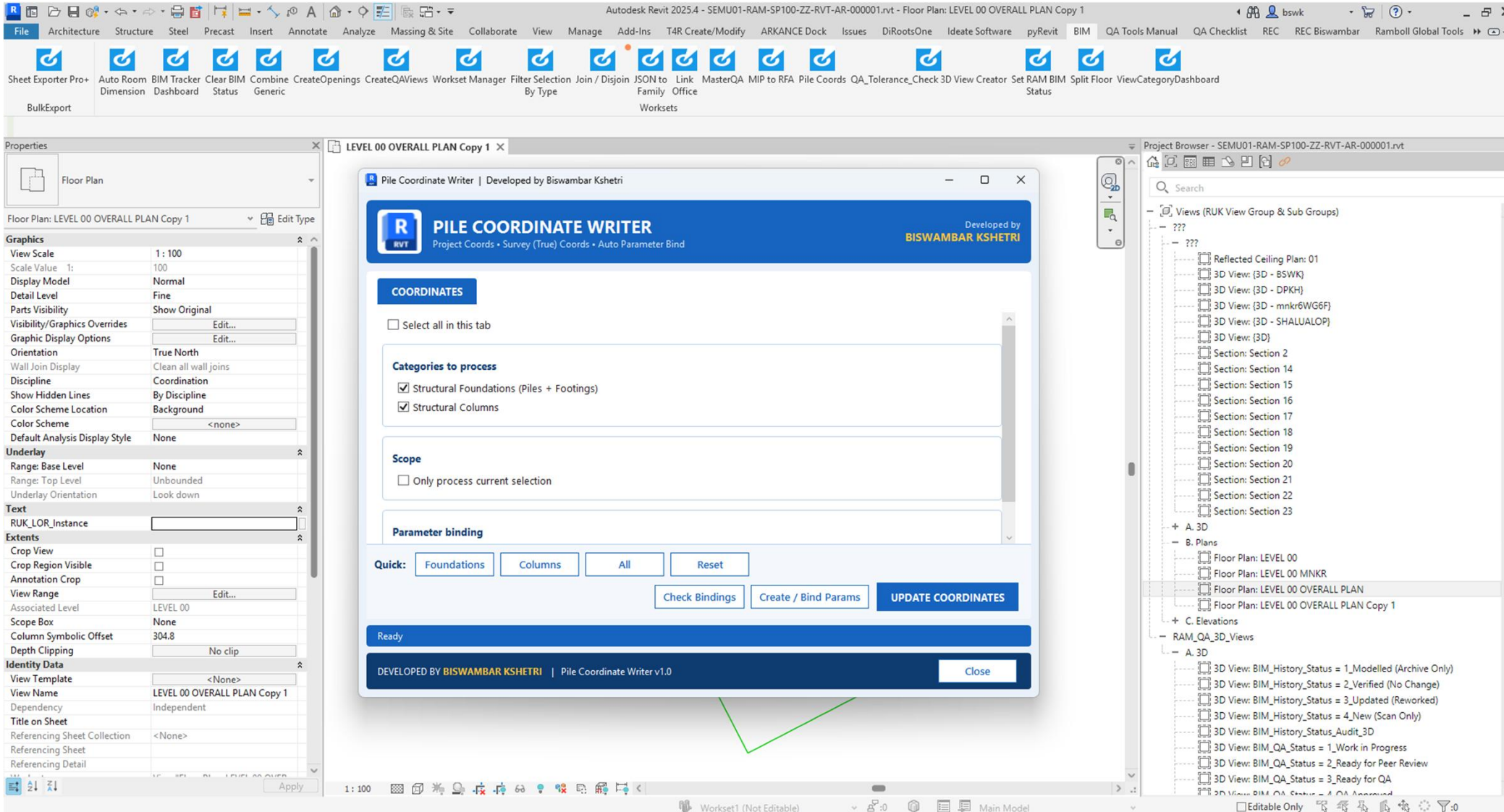
Worksets Manager

Create Worksets , Rename , Delete and Assign element into worksets



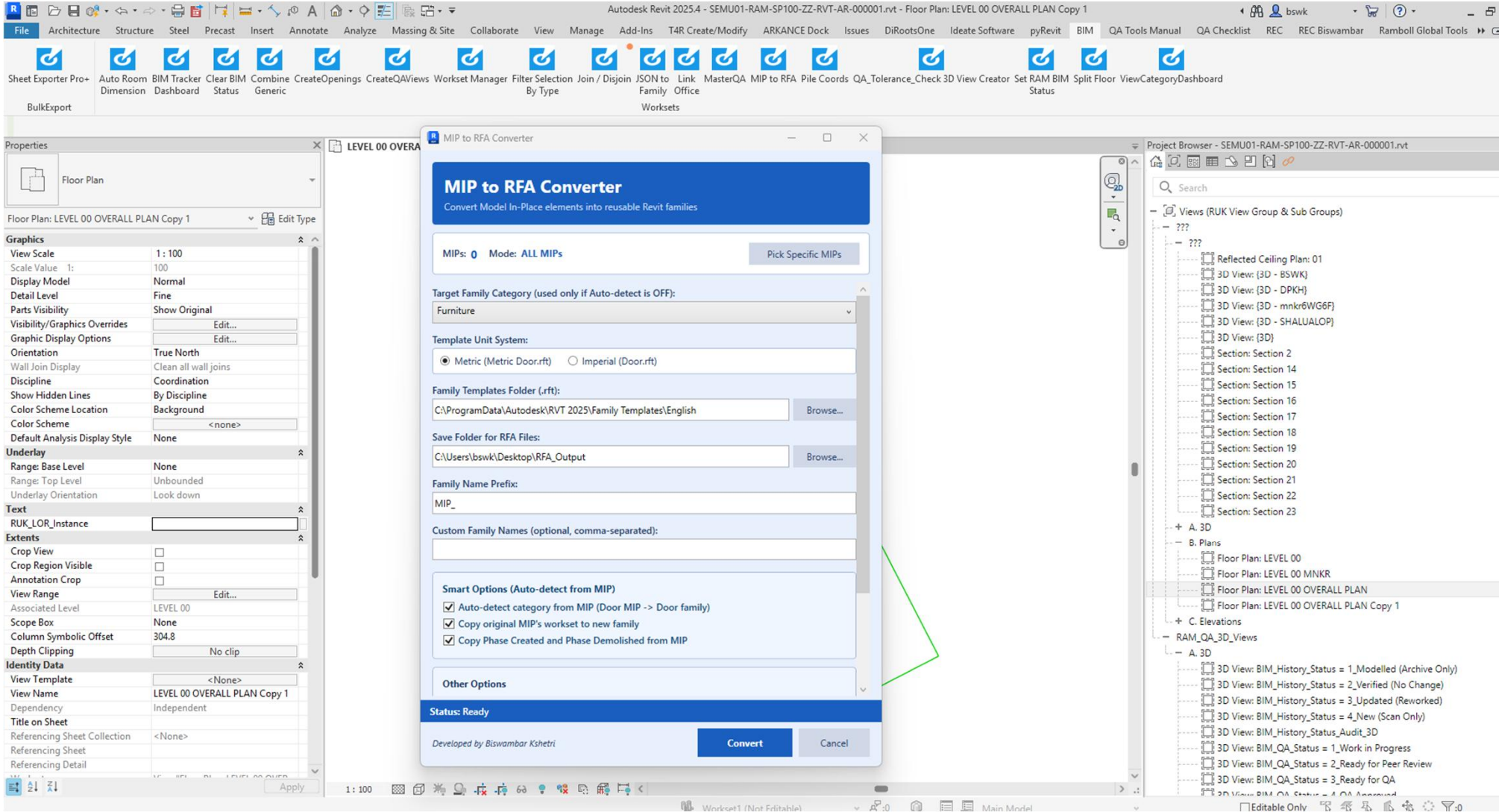
Pile and Column Coordinate Writer

Write XYZ Coordinate in Parameter in Revit Pile and Column



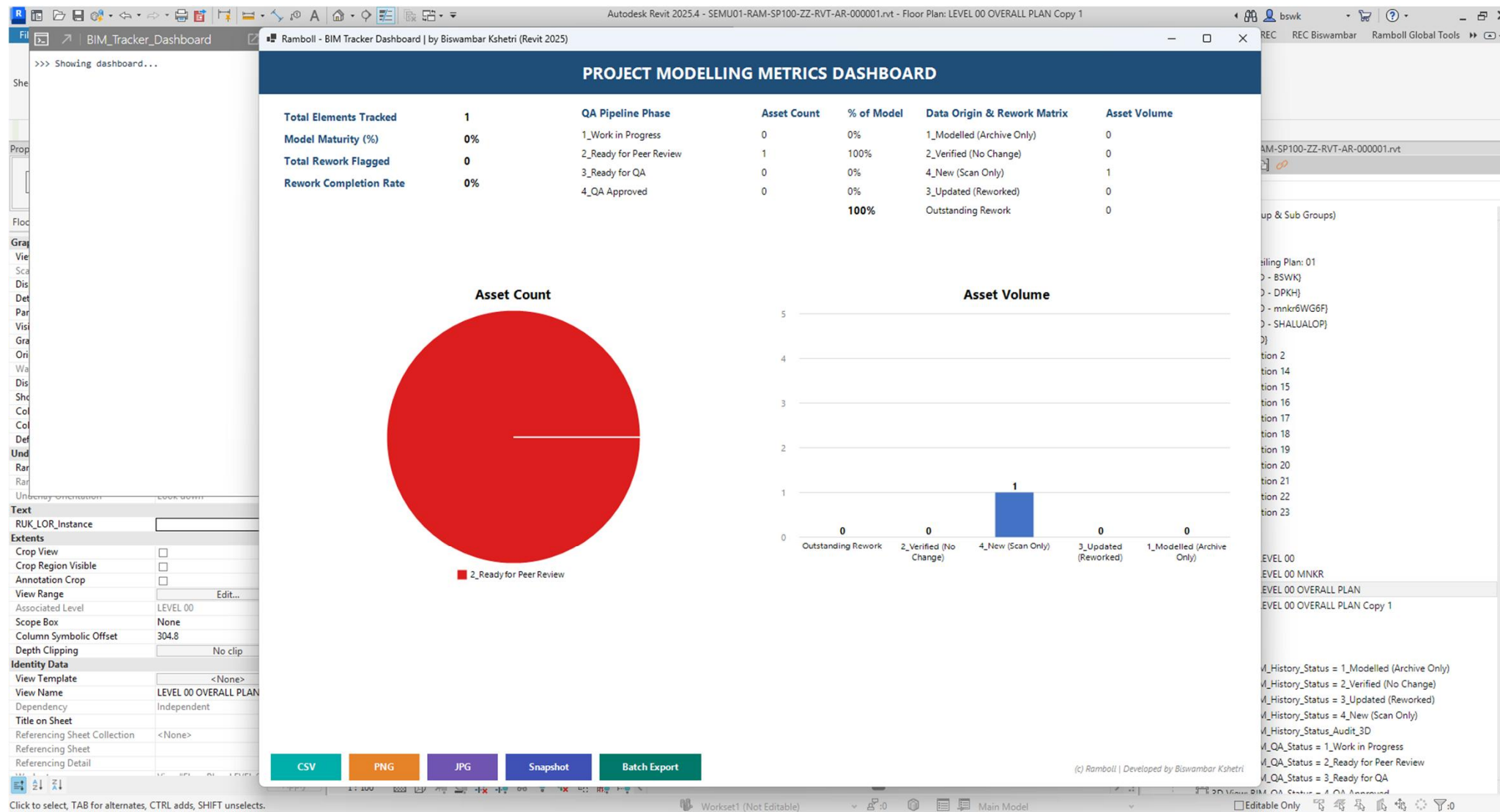
Covert Model in place to Generic

Convert Model in place to Generic Family and Place same location



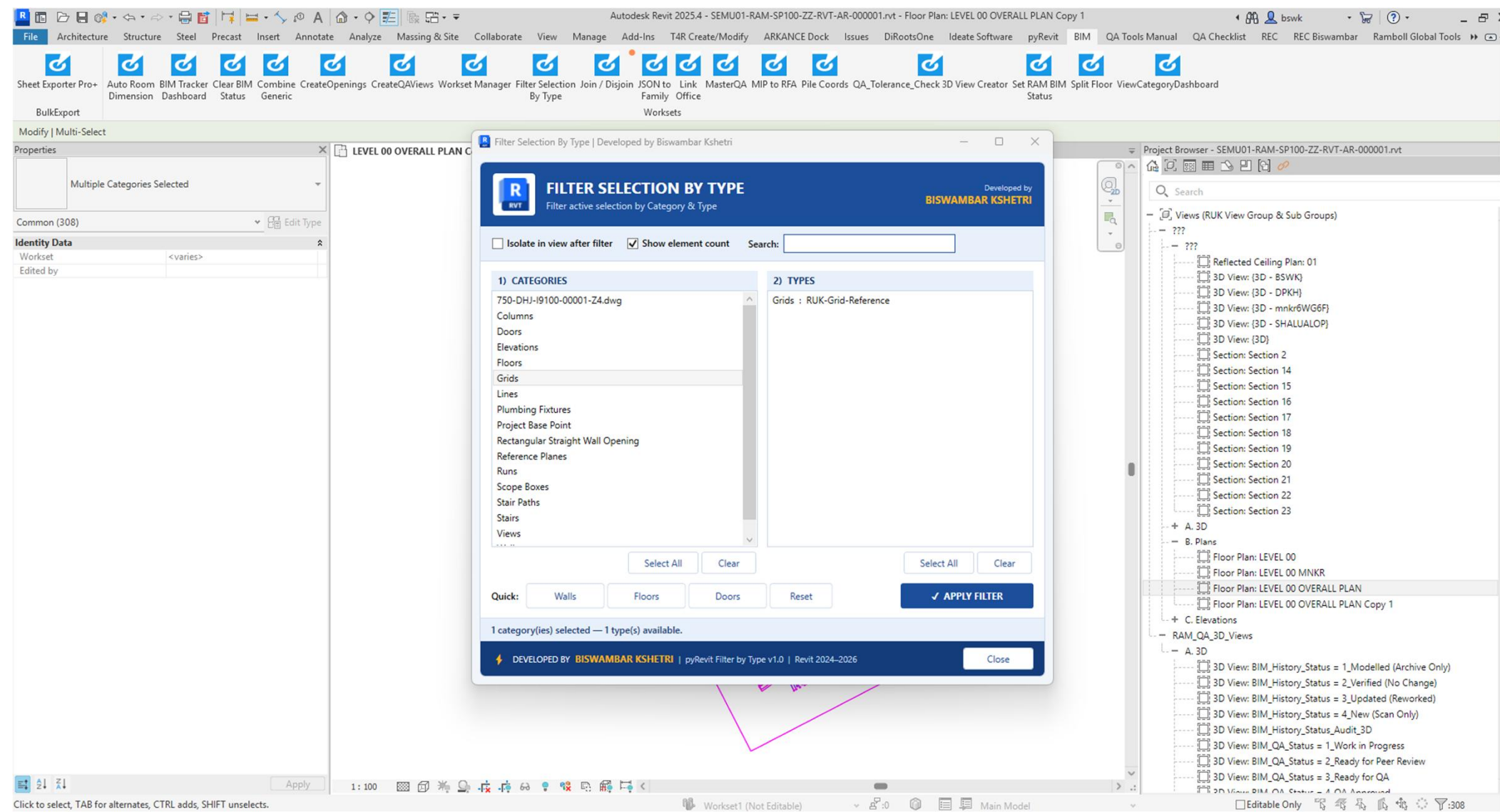
BIM Tracker Dashboard

This Tools Project Specific -St Pancras International Scan To BIM



Selection Filter by Type

Filter Element by Type

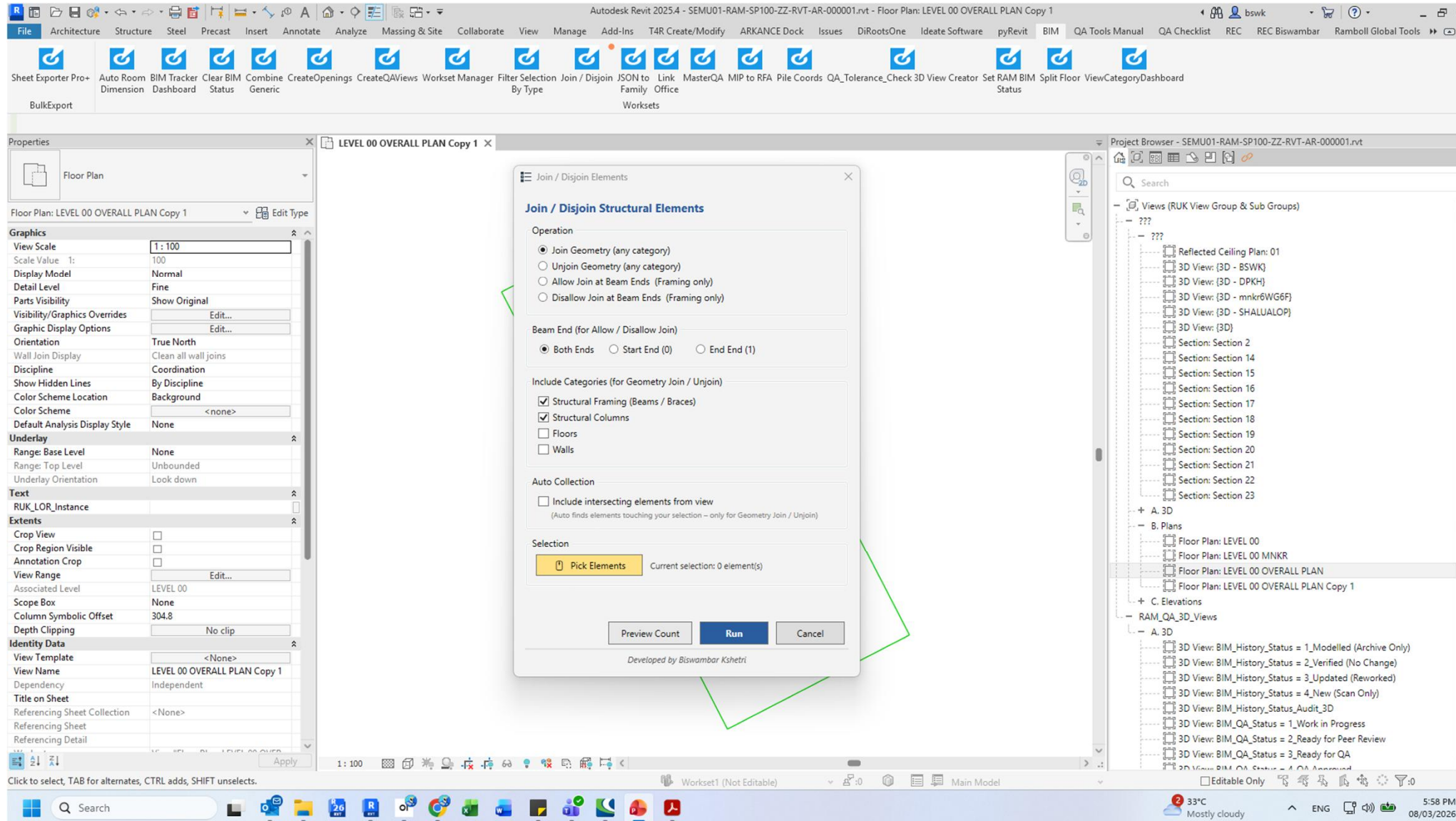


This Tools Project Specific -St Pancras International Scan To BIM



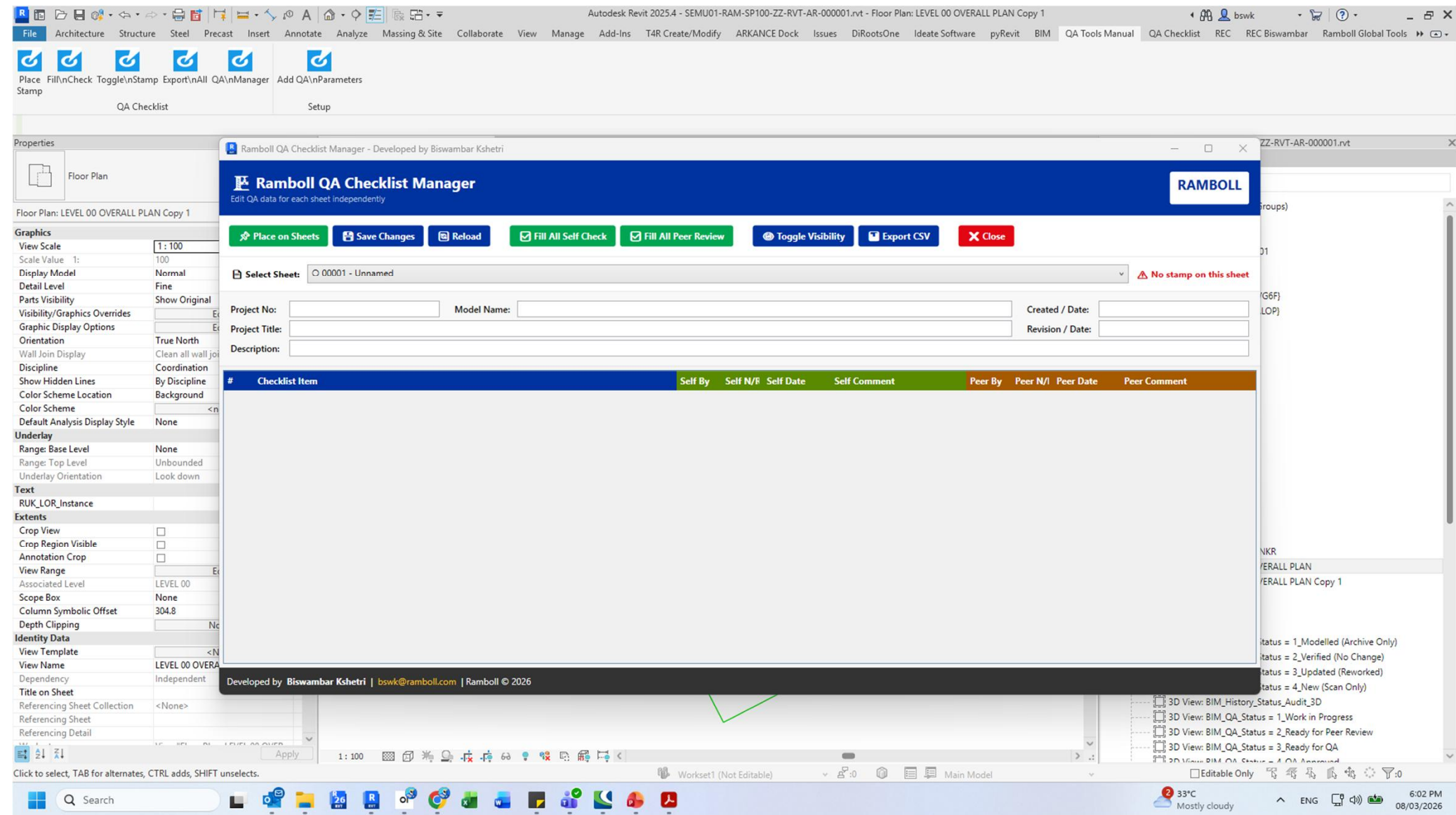
Join/Disjoin

Join and Disjoin Wall end and Structural Framing



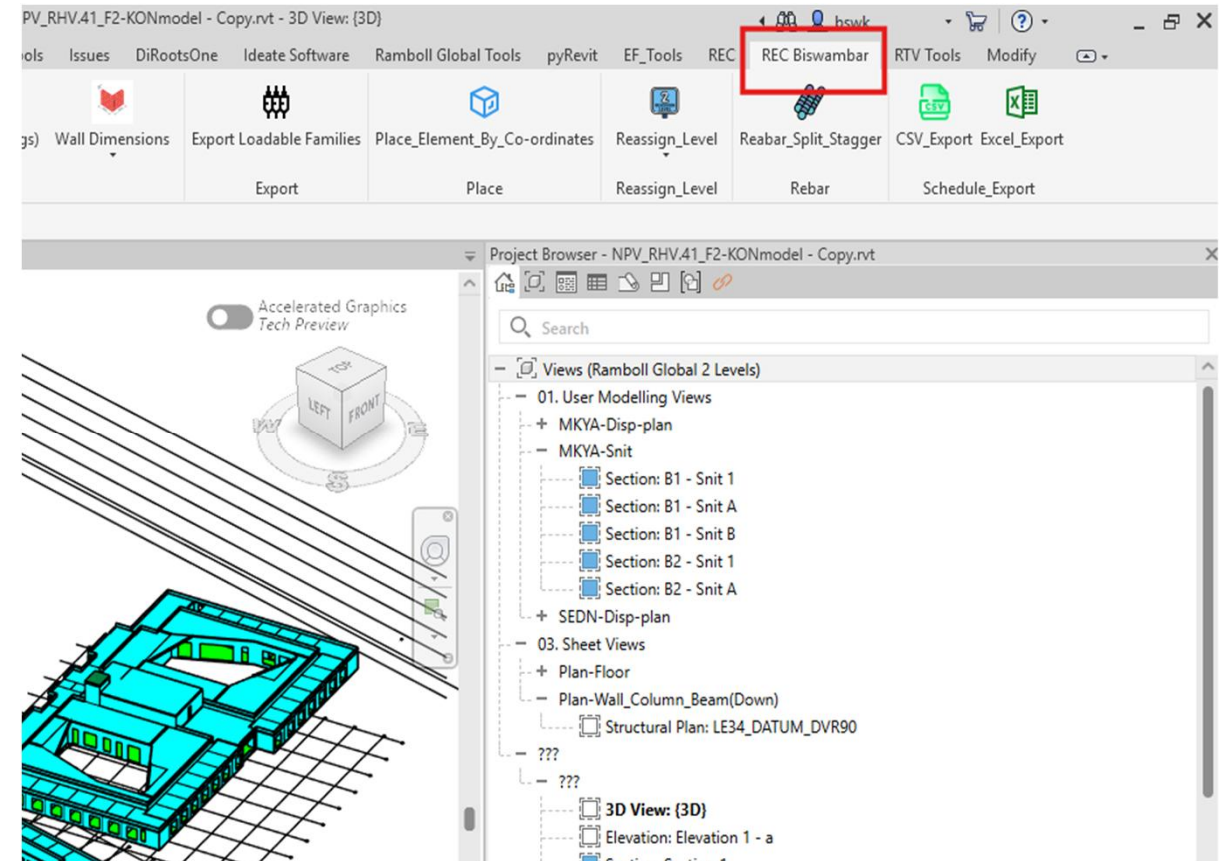
QA Checklist Manual in Sheet

Place Manual Checklist in Sheet for Manual Check



Why Use REC Customized User Tab?

- Time savings: Up to 60–70% on dimensioning tasks
- Consistency: Standardized output across projects
- Error reduction: Automated placement & checks
- Scalability: Built on pyRevit — easy to extend
- Team enablement: Empowers users to focus on design, not drafting



Where It Adds Value

- Structural projects with complex grids
- Foundation & pile layout drawings
- Project handover & family library creation
- QA / model audits before milestones

What's Next?

- Integration with Ramboll Global Tools
- Additional dimensioning automation for MEP
- Cloud-based settings sync across teams
- User feedback channel for new feature requests

Thank You

Questions & Discussion

Biswambar Kshetri
BSWK@ramboll.com
Ramboll – REC Building Gurgaon